

Using Natural Language Processing in a job interview to judge employer's required skills from employees

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Abstract—Traditional recruitment process comes with high operation costs and human bias. Due to corporations shift towards Industry 4.0, use of automated machines scoring has increased due to capabilities of interpreting qualitative data. This research develops and validates an interview prototype containing BART-MNLI Zero Shot Classifier installation to score open responses based on Big Five personality traits. Pearson Correlation was utilized to validate by comparing AI scores and human recruit scores. Results found a moderate positive relation (0.5286) and a 99 percent decrease in processing time compared to manual scoring. However, error analysis reveals lack of understanding in terms regarding industry-specific terms or technical acronyms. Findings suggest tuning on large language model's datasets to improve scoring accuracy across industry recruitment sectors.

Index Terms—Personality, traits, zero shot classifier, scoring

I. INTRODUCTION

This research discusses use of Natural Language Processing in a Human Computer Interaction-based prototype. Traditional interview are prone to bias and expenses to run an interview. With use of BART-MNLI, it prevents lacking qualitative depth of conversation while employing automated scoring based on required labels from open-ended answers.

This research will use a positivist approach since the algorithm scores the user's answers based on whether the response meets the employer's required personality traits or skills. The study will follow a deductive approach, aiming to testing the hypothesis that artificial intelligence (AI) can accurately match human-defined traits. This is a case study that focuses on validation of development and testing of interview prototype that uses BART-MNLI, which aims to score the answer based on labels (employer's required personality traits). It uses quantitative mono-method since it will focus on numerical score calculated by NLP model. This research will be conducted in cross-sectional where data is collected at a single point in time. Data will be collected via automated content analysis via BART-MNLI.

Due to shift towards "Industry 4.0", companies were forced to seek automated solutions for hiring. With rise of Transformer-based models using Natural Language Inference

(NLI), it allows machines to understand the intent behind a candidate's story. The aim of this research was to develop and validate an interview prototype utilizing Zero-Shot NLP to provide real-time scoring of candidates personality traits.

Traditional interview are prone to bias and expenses to run an interview. With use of BART-MNLI, it prevents lacking qualitative depth of conversation while employing automated scoring based on required labels from open-ended answers.

It is hypothesised that there is a significant positive correlation between the semantic scores generated by the BART-MNLI model and the evaluation scores provided by expert human recruiters for the same set of open-ended responses.

Research is aimed to develop and validate an interview prototype utilizing Zero-Shot NLP to provide real-time scoring of candidates personality traits. The purpose of this study is to display the effectiveness of Zero-Resource tool evaluating open-ended interview answers leading to reduction in recruiter bias.

II. LITERATURE REVIEW

Give a **brief** overview of the theme, history and facts. You should use the present tense since what you are documenting is current knowledge. Do not dedicate a lot to historical facts about the subject matter, but limit to a paragraph. Make sure you quantify by quoting statistics where relevant. Following is an example for a research in "Automatic cyber-bullying detection": *With the popularity of social media, predominantly amongst the adolescent and young adult generations, there appears to be a correlation between the increase in suicide rates and social networking adoption [?], with suicide attempts increasing by 10% amongst adolescents and by 5.6% amongst young adults for the period 2013-2017.*

Due to shift towards "Industry 4.0", companies were forced to seek automated solutions for hiring. With rise of Transformer-based models using Natural Language Inference (NLI), it allows machines to understand the intent behind a candidate's story. Candidate's personality scoring relied on interviews, although unstructured interviews are known to be 20 to 30 percent predictable due to human error and human bias

[1]. Since Natural Language processing has evolved, Large Language Models being the norm in every industry enables machines to score human personality traits from open ended response accurately. Literature review analyses use, performance, and accuracy of large language models in personality assesment

A. Pansare, P. Panwar, and P. Kosamkar [1] conducted a comparative study on natural language processing systems based on predicting personalities. Researchers used an MBTI dataset containing 8670 rows (personality set and fifty recent tweets) and two columns where it represents mostly all personality types. All tweets collected and questionnaire was used to judge the participants personality. The system, developed by researchers, consists of Logistic Regression, Random Forest Algorithm, Support Vector Machines, and XGBoost. Results found that the system scored high accuracy scores with thinking/feeling attribute being the highest, but lowest being perceiving vs judging.

S. R. Karra, S. T Nguyen, and T. Tulabandhula [2] conducted a comparative study from large language models on judging participants personality based on answers with use of Big Five Factors (Extraversion, Neuroticism, Agreeableness, Conscientiousness, Openness). Researchers used labels (each label has two ends such as from extroversion to introversion) on zero shot classifiers (ZSC), which are large language models, to judge the participants answers. Researchers used pre-processed datasets, from BookCorpus and Wikipedia, containing both sentences and small paragraphs for the ZSC to judge on.

S. Sikström, I. Valavičiūtė, and P. Kajonius [3] analysed the validity of NLP judging personality based on word-based responses. 663 participants were recruited from United States of America, United Kingdom, and other countries with use of Prolific.io which is an online recruitment tool to collect data. The session consists of two phases. Phase 1 was where participants wrote a description about own or someone else's specific personality trait with use of keywords and no mention of personality trait. Phase 2 was where other participants read participant's written description. IPIP-NEO-30, derived from International Personality Item Pool, was used to measure, measured by participants, five personality factors, extraversion, openness, agreeableness, conscientiousness, and neuroticism. Demographic information such as age, gender, birth of country, education and confidence level were collected with use of questionnaires. The user-made descriptions, written in Microsoft Word, were analysed with use of GPT-4 and BERT based on the big five personality traits. Results found higher prediction accuracy than the rating scale, higher reliability, and better observation than self-report.

J. Kurek et al. [4] conducted comparative study on zero shot classifier as a recruitment tool. Dataset contained unique job candidates, unique jobs, and matches between jobs and candidates totalling up to 3169 records. Only English documents were used only used due to researchers focus on English documents only. Models such as all-MiniLM-L2-v2 and LaBSe were selected by the researchers due to "mod-

els' ability to produce embedding vector representations that accurately capture semantic meanings". Results found high accuracy scores, importance of selecting the appropriate model as a recruitment tool, high performance, and recommended modifications on pre-trained models.

L. Hu et al. [5] made use of large language model (LLM) to analyse personality based on user's input. Researchers collected user's posts (from Kaggle and Pandora) which was used on LLM, GPT 3.5 Turbo, to analyse from three perspectives (semantic, sentiment, and linguistic perspective). Prompts were made to LLM to generate explanation on each MBTI trait and calculate similarity between the user and the label description with use of soft labels. Results found that the model, constructed by the researchers, outperformed all traditional methods that detect personalities, performance impact from focus on linguistic over semantic, and struggles to generate explanations with use of LLMs rather than specific small models. Limitations include limited dataset, and input length.

All sources touched on use of LLMs and NLPs in the context of job recruitment. Referenced studies made use of GPT-4, BERT, and BART as zero shot classifiers, and Logistic Regression, Random Forest, and XGBoost to predict personality traits from social media datasets. Pansare et al. [1] made use of specific models such as XGBoost, and Random Forest resulted in high accuracy but required large and pre-labelled datasets. On the other hand, Karra et al. [2] and Kurek et al. [4] employed Zero-shot classifiers showed that personality scoring was able to be executed without specific training, supported by Sikstrom et al. [3] who found that GPT-4 offered high reliability than self-report scales. Although LLMs outperform traditional methods, LLMs often struggle to generate explanation based on traits compared to smaller specific models. In addition, Kurek et al. [4] identified a linguistic gap due to limiting found documents to only English, as well as Hu et al. finding performance gaps when focus shifts to linguistic perspectives over semantics. Limitations found in this literature review represent a knowledge gap that the research gap needed to be addressed in this research by using BART-MNLI to evaluate and score open-ended candidate stories.

Finally, sources reviewed in the literature review were peer-reviewed and contained in-depth analysis, differed from non-academic sources that lacks evidential findings due to lack of analysis and use of biased sources. By putting focus on academic work, this literature review identified technical gaps such as linguistic constraints and bias driven by pretrained data. However, non-academic material would only provide benefits of use of AI, but do not provide any findings based on limitations found in academic work.

III. RESEARCH METHODOLOGY

The zero shot classifier will use the big five personality framework to score based on user's open response. A questions related to Big-Five personality framework were asked to users.

TABLE I
COMPARATIVE ANALYSIS OF LITERATURE: DATA, ALGORITHMS, AND METRICS

Source	Data Set Type	Algorithm/Model	Primary Metric
Pansare [1]	Social Media (MBTI)	XGBoost / Random Forest	Accuracy
Karra [2]	Wikipedia/Corpus	Zero-Shot (LLM)	Semantic Mapping
Sikström [3]	Prolific.io Narratives	GPT-4 / BERT	Pearson Correlation
Kurek [4]	Job-Matching Data	all-MiniLM / LaBSE	Matching Score
Hu [5]	Kaggle/Pandora Posts	GPT-3.5 Turbo	Soft Labels

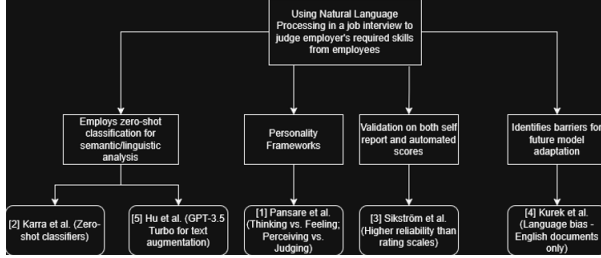


Fig. 1. Literature Map

User's input was collected and scored with a specific big-five trait. Data would be collected based on scores scored by the ZSC and time taken to score, A comparative analysis would be conducted scores both by ZSC and human reviewers. A Pearson Correlation would be used to measure strength between AI and human scores.

Traditional interview are prone to bias and expenses to run an interview. With use of BART-MNLI, it prevents lacking qualitative depth of conversation while employing automated scoring based on required labels from open-ended answers. The research hypothesized that *there is a significant positive correlation between the semantic scores generated by the BART-MNLI model and the evaluation scores provided by expert human recruiters for the same set of open-ended responses.*

The primary aim of this reserach was to develop and make use of a simple python app that utilized Zero-Shot Natrual Language Processing to display real-time assesment of candidate personality traits based on open-ended responses. Our research objectives were:

- 1) Implement BART-MNLI to generate a score for each user response
- 2) Conduct correlation analysis between AI scores and human scores
- 3) Measure and compare time taken to process automated scoring against time taken by human recruiters to evaluate users answers
- 4) Conduct error analysis on instances with low correlation to detect linguistic issues or terms where ZSC fail to score.

The research aimed to answer the following research questions:

- 1) To what extent does the BART-MNLI zero-shot classification model correlate with expert human recruiter evaluations when scoring specific employer-required skills in open-ended interview responses?

- 2) Does the use of automated NLI-based scoring significantly reduce the time required for the initial candidate screening phase compared to traditional human-led qualitative analysis?
- 3) What are the primary linguistic limitations of the BART-MNLI model when interpreting nuanced or industry-specific terminology in candidate "stories"?

Finally, you need a plan on how you intend to answer the research questions which will help you prove/disprove the hypothesis. Different research areas would require specific pipelines, yet a generic one is shown in Fig 2. After presenting an illustration of your pipeline it is recommended to documenting what has been done at every stage. Remember you should use past tense in this section since you are document what you have already done, since this paper will be published at the end of your research.



Fig. 2. Research Pipeline

A. Research Onion

This research used a positivist approach since the algorithm scored the user's answers based on whether the response met the employer's required personality traits or skills.

The study followed a deductive approach, aimed to test the hypothesis that artificial intelligence (AI) could accurately match human-defined traits.

This was a case study that focused on validation of development and testing of interview prototype that used BART-MNLI, which aimed to score the answer based on labels (employer's required personality traits). It used quantitative mono-method since it would focus on numerical score calculated by NLP model. This research was conducted in cross-sectional where data was collected at a single point in time. Data was collected via automated content analysis via BART-MNLI.

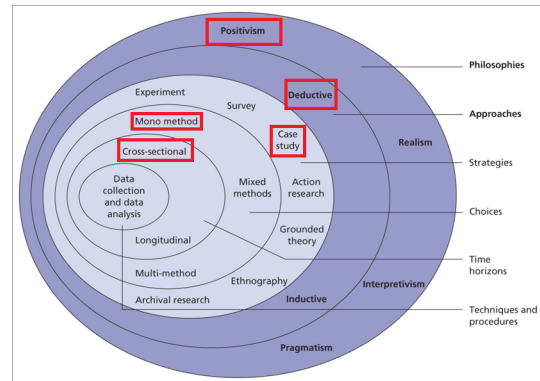


Fig. 3. Research Onion

B. Validity, Reliability, and Generalisability/Transferability

BART-MNLI's potential role in personality assesment was validated by comparing scores with datasets between AI and human reviewers. A fixed random seed for BART-MNLI was used to validate inputs that always generate identical scores.

Since BART-MNLI is a ZSC, it was designed to score based on labels specified without training. However, since the dataset is trained on general sources such as wikipedia, the dataset of user responses may contain industrial-specific terms that are not understood by ZSC.

C. Ethical Considerations

BART-MNLI is a pre trained from massive datasets such as Wikipedia which would include misinformation as well as morally wrong information. Due to bias, this could lead to a possibility of unfair scoring based on linguistic style. All sensitive information would be removed for the participants to remain anonymous. Finally, ZSC are known to give lack of reasoning behind scoring based on user response which leads to lack of transparency.

IV. FINDINGS & DISCUSSION OF RESULTS

A. AI Score vs. Recruiter Score Correlation

After responses from 4 participants, both AI and human reviewer scores calculate a pearson correlation of 0.5286 resulting in a moderate positive relationship as show in figure 4. The prototype was proven to be used as a supportive rather than replacing human reviewers.

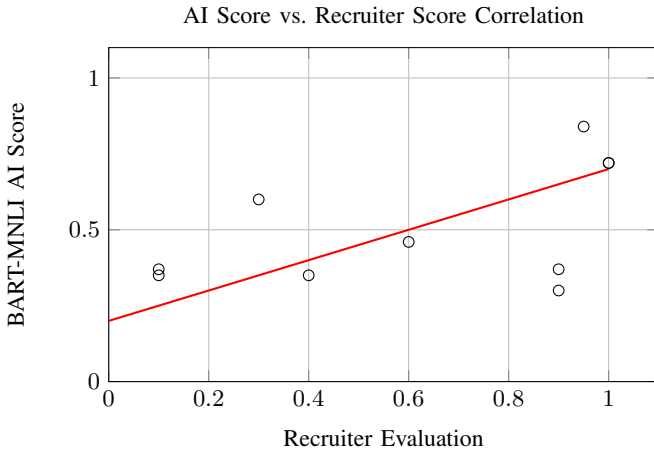


Fig. 4. Scatter plot illustrating the moderate positive correlation ($r = 0.5286$) between automated and manual scoring.

However, ZSC struggled to identify key terms found in user's open ended responses. For instance, the term 'SLA' was scored higher by the human recruiter than AI. BART-MNLI was trained on general sources such as wikipedia which possesses misunderstanding the industry context.

The results align with Sikström et. al [3] since the source stated that Natural Language Processing is sensitive to specific word choice such as technological terms. This confirmed

Sikström's theory where artificial intelligence scoring significantly change if some words are subtle.

In the session, model's confidence score dropped if the user's open response is shorter than 20 words resulting in a minimum character count.

TABLE II
COMPARATIVE ANALYSIS OF AI VS. RECRUITER SCORING FOR SELECTED RESPONSES

ID	Trait	Candidate Response	AI	Human
0	Cons.	I managed three simultaneous projects by using a Gantt chart...	0.72	1.00
1	Open.	I noticed our traditional filing system was inefficient...	0.30	0.90
2	Extra.	I hate talking to people I don't know. I'd much rather sit in the back...	0.30	0.90
3	Agree.	I don't care about making friends at work. If I think an idea is stupid, I'll say it...	0.30	0.90
4	Neuro.	When confronted with technical debt, I focus on root-cause analysis without escalating...	0.30	0.90

TABLE III
OPERATIONAL EFFICIENCY COMPARISON: AI VS. MANUAL EVALUATION

Metric	BART-MNLI Prototype	Human Recruiter
Total Evaluations	20	20
Avg. Time per Response	0.35 seconds	~60.0 seconds
Total Process Time	7.04 seconds	~20 minutes
Time Reduction	99.4%	-

V. CONCLUSION

Results found a moderate positive correlation of 0.5286 which is evidenced that AI can be used in a personality assesment. However, ZSC is shown that is suitably used as a support tool rather than a complete human recruiter replacement. Reduction on time when scoring open ended responses based on a specific Big-Five trait is massive. AI scoring was executed under a second when evaluating user's open ended response, while a human recruiter took 1-2 minutes to evaluate and score. Despite the Pearson Correlation score, it was also found AI's inability to identify linguistic and industry-specific terms which influenced AI's scoring. For instance, AI failed to recognise terms such as "SLA" and "CRM" in one of the user's repsonse.

A primary limitation identified was small sample size which consisted of 4 users. Due to the sample size, this might have changed the validation of the prototype's use in the persoanlity assesment in the context of job recruitment. As previously stated, ZSC lacks understading in industrial specifics due to model's general training sources. 1 human recruiter resulted in possible human bias which impacted human recruiter's scoring. In addition, BART-MNLI was only comapred in context of AI scoring which negatively impacts validation of AI's role in personality assesment.

Models should be fine tuned in order to improve identifying specific industrial terms such as HR-specific datasets. Alternatively, industry-specific models should be used for personality assessment to combat technical jargon. Finally, sample size should be increased by inviting more participants or make use of datasets that contains user responses used for personality assessment.

APPENDIX A SUPPORTING MATERIAL

You can add screen shots and statistics. Stick to essential information.

ACKNOWLEDGEMENT

You can dedicate this section for special assistance that you were given by a 3rd party. Not your mentor, relatives or questionnaire participants.

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